Vision-Language Model Comparison

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Contents

1 Introduction and Motivation

2 State of the Art

3 Objectives

4 Planning

5 Costs

6 Experimental Methodology

6.1 Model Selection

6.1.1 The Models Chosen

As mentioned in the report, the models chosen are:

Proprietary/API models

- `gpt-5.5`
- `gpt-5.4`
- `o3`
- `o4-mini`
- `claude-opus-4-8`
- `claude-opus-4-7`
- `claude-opus-4-6`
- `claude-sonnet-4-6`
- `claude-sonnet-4-5`
- `gemini-3.1-pro-preview`
- `gemini-3.5-flash`
- `qwen3.7-plus-preview`
- `qwen3.6-plus`

Open-weight models

- `qwen3.6-27b`
- `qwen3.5-397b-a17b`
- `qwen3-vl-235b-a22b`
- `kimi-k2.6`
- `kimi-k2.5-thinking`
- `gemma-4-31b`

- `gemma-4-26b-a4b`
- `mimo-v2.5`
- `mimo-v2.5-pro`
- `mistral-medium-3.5`
- `mistral-small-4`
- `dots.vlm1`
- `dots.ocr`
- `glm-4.5v-thinking`
- `glm-4.1v-thinking`
- `skywork-r1v3-38b`
- `skywork-r1v2-38b`

6.1.2 The Procedure

1. Freeze the four leaderboard snapshots: Arena Vision proprietary, Arena Vision open-weight, MMMU-Pro proprietary, and MMMU-Pro open-weight.
2. For each leaderboard, identify the five highest-ranked model families by scanning the ranking from top to bottom. Skip models that cannot be evaluated through a generally available API or downloadable checkpoint. Claude Opus and Claude Sonnet are treated as distinct families because they occupy different capability, cost, and deployment tiers.
3. For each selected family, choose two models:
   (a) the highest-ranked model from that family in the leaderboard. If this model is no longer available through its provider, replace it with the closest currently available model from the same family. For example, Gemini 3.0 Pro appears in the leaderboard, but it is no longer available in Google AI Studio; it is therefore replaced by Gemini 3.1 Pro Preview, accessed through the endpoint `gemini-3.1-pro-preview`;
   (b) the strongest currently available distinct model from the same family. If the strongest current model coincides with the leaderboard-selected model, choose the next strongest currently available distinct model from that family.
4. Collapse variants differing only by inference or serving mode, such as `thinking`, `high`, `instant`, or `flash`, keeping the strongest or highest-ranked variant unless the distinction is itself relevant to the experiment.
5. Remove duplicates across the four leaderboards and evaluate each remaining model once.

This procedure gives at least 10 models.

When a provider exposes a model only through a preview endpoint, the preview endpoint is treated as the implementation of the corresponding leaderboard model, provided that it is the closest available version of the same model family. For example, a leaderboard entry written as Gemini 3.1 Pro is evaluated using `gemini-3.1-pro-preview`.

6.1.3 The Leaderboards

Model name	Prop. rank	Overall rank	Score	Date	Family	Company
GPT-5.4 Thinking w/ tools	1	1	82.1	2026-03-05	GPT-5.4	OpenAI
GPT-5.4 Thinking w/o tools	2	2	81.2	2026-03-05	GPT-5.4	OpenAI
Gemini 3.0 Pro	3	3	81.0	2025-11-18	Gemini 3	Google DeepMind
Gemini 3.1 Pro Thinking (High)	4	4	80.5	2026-02-19	Gemini 3.1	Google DeepMind
Muse Spark Thinking	5	5	80.4	2026-04-08	Muse Spark	Meta
GPT-5.2 Thinking w/o Python	5	5	80.4	2025-12-11	GPT-5.2	OpenAI
GPT-5.2 Thinking w/o tools	7	7	79.5	2025-12-11	GPT-5.2	OpenAI
GPT-5.1 Thinking	8	8	79.0	2025-11-13	GPT-5.1	OpenAI
GPT-5 w/ thinking	9	9	78.4	2025-08-07	GPT-5	OpenAI
Claude Opus 4.6 w/ tools	10	10	77.3	2026-02-05	Claude Opus 4.6	Anthropic
o3	11	12	76.4	2025-04-16	o3	OpenAI
GPT-5.1	12	13	76.0	2025-11-13	GPT-5.1	OpenAI
Claude Sonnet 4.6 w/ tools	13	14	75.6	2026-02-17	Claude Sonnet 4.6	Anthropic
Claude Sonnet 4.6 w/o tools	14	15	74.5	2026-02-17	Claude Sonnet 4.6	Anthropic
Claude Opus 4.6 w/o tools	15	16	73.9	2026-02-05	Claude Opus 4.6	Anthropic
Claude Opus 4.5	15	16	73.9	2025-11-24	Claude Opus 4.5	Anthropic
Claude Sonnet 4.5	17	20	68.9	2025-09-29	Claude Sonnet 4.5	Anthropic
Gemini 2.5 Pro 05-06	18	22	68.0	2025-05-06	Gemini 2.5	Google DeepMind
Seed 1.5-VL Thinking	19	23	67.6	2025-05-13	Seed 1.5-VL	ByteDance

Table 1: Highest-ranked proprietary/API models on the MMMU-Pro leaderboard snapshot [38], extended until five owning companies are represented. Ranks are model-only ranks, excluding human baselines. Tied scores share the same rank.

Model name	Open-weight rank	Overall rank	Score	Date	Family	Company
Gemma 4 31B	1	11	76.9	2026-04-02	Gemma 4	Google DeepMind
Gemma 4 26B A4B	2	18	73.8	2026-04-02	Gemma 4	Google DeepMind
dots.vlm1	3	19	70.1	2025-08-05	dots.vlm	RedNote
Qwen3-VL 235B-A22B	4	21	68.1	2025-09-22	Qwen3-VL	Alibaba Cloud
GLM-4.5V w/ Thinking	5	25	65.2	2025-08-11	GLM-4.5V	Z.ai / Zhipu AI
GLM-4.1V w/ Thinking	6	29	57.1	2025-07-01	GLM-4.1V	Z.ai / Zhipu AI
Skywork-R1V3-38B	7	30	55.4	2025-07-05	Skywork-R1V	Skywork AI

Table 2: Highest-ranked open-weight models on MMMU-Pro [38], extended until five distinct owning companies are represented. Ranks are model-only ranks, excluding human baselines. Seed models are excluded because they are treated here as API-accessible rather than open-weight.

Model name	Prop. rank	Overall rank	Score	Date	Family	Company
claude-opus-4-7-thinking	1	1	1308 ± 8	2026-04-16	Claude Opus 4.7	Anthropic
claude-opus-4-6-thinking	2	2	1303 ± 8	2026-02-05	Claude Opus 4.6	Anthropic
claude-opus-4-7	3	3	1300 ± 8	2026-04-16	Claude Opus 4.7	Anthropic
claude-opus-4-6	4	4	1296 ± 7	2026-02-05	Claude Opus 4.6	Anthropic
muse-spark	5	5	1295 ± 10	2026-04-08	Muse Spark	Meta
claude-opus-4-8-thinking	6	6	1294 ± 13	2026-05-28	Claude Opus 4.8	Anthropic
gemini-3-pro	7	7	1289 ± 8	2025-11-18	Gemini 3	Google
gpt-5.5	8	8	1286 ± 9	2026-04-23	GPT-5.5	OpenAI
gpt-5.4-high	9	9	1283 ± 8	2026-03-05	GPT-5.4	OpenAI
gpt-5.5-high	10	10	1282 ± 9	2026-04-23	GPT-5.5	OpenAI
gpt-5.2-chat-latest-20260210	11	11	1279 ± 7	2025-12-11	GPT-5.2	OpenAI
claude-opus-4-8	12	12	1279 ± 12	2026-05-28	Claude Opus 4.8	Anthropic
gemini-3.1-pro-preview	13	13	1278 ± 6	2026-02-19	Gemini 3.1	Google
claude-sonnet-4-6	14	14	1278 ± 7	2026-02-17	Claude Sonnet 4.6	Anthropic
gpt-5.4	15	15	1275 ± 8	2026-03-05	GPT-5.4	OpenAI
gpt-5.5-instant	16	16	1275 ± 9	2026-04-23	GPT-5.5	OpenAI
gemini-3-flash	17	17	1271 ± 6	2025-12-17	Gemini 3	Google
qwen3.7-plus-preview	18	18	1263 ± 11	2026-05-19	Qwen3.7	Alibaba

Table 3: Highest-ranked proprietary models on Arena Vision Overall [39, 40], extended until five distinct companies are represented.

Model name	Open rank	Overall rank	Score	Date	Family	License	Company
kimi-k2.6	1	19	1261 ± 8	2026-04-20	Kimi K2.6	Modified MIT	Moonshot
gemma-4-31b	2	24	1253 ± 7	2026-04-02	Gemma 4	Apache 2.0	Google
qwen3.5-397b-a17b	3	26	1249 ± 7	2026-02-16	Qwen3.5	Apache 2.0	Alibaba
kimi-k2.5-thinking	4	28	1247 ± 7	2026-01-27	Kimi K2.5	Modified MIT	Moonshot
kimi-k2.5-instant	5	34	1238 ± 11	2026-01-27	Kimi K2.5	Modified MIT	Moonshot
mimo-v2.5	6	35	1238 ± 8	2026-04-27	MiMo-V2.5	MIT	Xiaomi
gemma-4-26b-a4b	7	36	1238 ± 8	2026-04-02	Gemma 4	Apache 2.0	Google
qwen3.5-122b-a10b	8	41	1226 ± 7	2026-02-24	Qwen3.5	Apache 2.0	Alibaba
qwen3.5-27b	9	45	1220 ± 7	2026-02-24	Qwen3.5	Apache 2.0	Alibaba
qwen3-vl-235b-a22b-instruct	10	49	1215 ± 7	2025-09-22	Qwen3-VL	Apache 2.0	Alibaba

Table 4: Highest-ranked open-weight models visible in the provided Arena Vision Overall excerpt [39, 40]. The excerpt reaches four distinct companies: Moonshot, Google, Alibaba, and Xiaomi.

6.1.4 Models Selected per Leaderboard

In the lists below, model names refer to the model identifiers or closest currently available implementation used in the experiments. When a leaderboard entry refers to an unavailable or superseded model, the closest currently available model from the same family is used instead. For example, Gemini Pro entries are evaluated through `gemini-3.1-pro-preview`. Similarly, labels such as `thinking`, `high`, or `w/ tools` are treated as inference or serving configurations rather than separate model families.

Models selected from Table 1. The five selected families are GPT, Gemini, Claude Opus, o-series, and Claude Sonnet. The models included from this leaderboard are:

- **GPT**: `gpt-5.4`; `gpt-5.5`.
- **Gemini**: `gemini-3.1-pro-preview`; `gemini-3.5-flash`.
- **Claude Opus**: `claude-opus-4-6`; `claude-opus-4-8`.
- **o-series**: `o3`; `o4-mini`.

- **Claude Sonnet**: `claude-sonnet-4-6`; `claude-sonnet-4-5`.

Muse Spark is excluded because its API is not generally available.

Models selected from Table 2. The five selected families are Gemma, dots, Qwen, GLM, and Skywork. The models included from this leaderboard are:

- **Gemma**: `gemma-4-31b`; `gemma-4-26b-a4b`.
- **dots**: `dots.vlm1`; `dots.ocr`.
- **Qwen**: `qwen3-vl-235b-a22b`; `qwen3.6-27b`.
- **GLM**: `glm-4.5v-thinking`; `glm-4.1v-thinking`.
- **Skywork**: `skywork-r1v3-38b`; `skywork-r1v2-38b`.

Models selected from Table 3. The five selected families are Claude Opus, Gemini, GPT, Qwen, and Claude Sonnet. The models included from this leaderboard are:

- **Claude Opus**: `claude-opus-4-7`; `claude-opus-4-8`.
- **Gemini**: `gemini-3.1-pro-preview`; `gemini-3.5-flash`.
- **GPT**: `gpt-5.5`; `gpt-5.4`.
- **Qwen**: `qwen3.7-plus-preview`; `qwen3.6-plus`.
- **Claude Sonnet**: `claude-sonnet-4-6`; `claude-sonnet-4-5`.

Muse Spark is excluded because its API is not generally available.

Models selected from Table 4. The visible excerpt contains four selected families: Kimi, Gemma, Qwen, and MiMo. The fifth family is obtained by extending the leaderboard beyond the visible excerpt. The models included from this leaderboard are:

- **Kimi**: `kimi-k2.6`; `kimi-k2.5-thinking`.
- **Gemma**: `gemma-4-31b`; `gemma-4-26b-a4b`.
- **Qwen**: `qwen3.5-397b-a17b`; `qwen3.6-27b`.
- **MiMo**: `mimo-v2.5`; `mimo-v2.5-pro`.
- **Mistral**: `mistral-medium-3.5`; `mistral-small-4`.

6.2 Dataset Selection

6.2.1 Dataset selection procedure

For each period {pre-2015, 2015, 2016, . . . , 2024, 2025}, I selected the three multimodal datasets with the highest number of citations on `Semantic Scholar` (`semanticscholar.org/`) as Google scholar does not allow me to order items by number of citations. Concretely, for each year bin I queried `Semantic Scholar` with the keywords `"dataset"`, `"image"`, and `"objects"` and then ranked results by the number of citations reported by the Semantic Scholar platform [37]. A dataset was included only if it satisfied all of the following:

1. **Search visibility.** The dataset must appear in `Semantic Scholar` search results for at least one of the keywords *dataset*, *image*, or *objects*.

2. **Multimodality with vision.** The dataset must be multimodal and include **images or videos**. Purely non-visual multimodal datasets are excluded (e.g., ESC (2015) is excluded because it focuses on audio). Datasets with 3D/geometry are acceptable **if** they also include 2D images or videos (e.g., FlyingThings3D (2015) qualifies); datasets lacking a 2D visual modality are excluded (e.g., Google Scanned Objects (2022) excluded).

3. **Scope exclusions.** The dataset must **not** be:
   - fully medical (e.g., CheXpert (2019) excluded),
   - exclusively aerial/remote-sensing imagery (e.g., EuroSAT (2017) excluded),
   - images or videos taken from vehicles' cameras (e.g., nuScenes (2019) excluded; HDR cameras on top of vehicles such as the ones used for the street view on Google Maps are OK, hence Cityscapes (2016) is included).

Applying these rules, I retained, for each period, the top three qualifying datasets by publication count.

6.2.2 Summary of the Datasets as a Whole

Abbreviations. Licenses: **FNCR** = free for non-commercial research; **FAP** = free for any purpose; **MRL** = Meta Research License; **OC** = original copyright; **Unk.** = unknown. Standard license identifiers such as CC BY, CC0, MIT, Apache-2.0, and GPL-3.0 are retained. Elements: **Img** = images; **Vid** = videos; **Lbl** = labels; **Act** = action labels; **Seg** = segmentation; **Cap** = captions; **BB** = bounding boxes; **Rel** = relationships; **QA** = question–answer pairs; **Aud** = audio; **Txt** = text; **Pr** = prompts; **Sc** = scores; **Pref** = preference. Evaluation types: **A** = answering questions; **B** = bounding box object detection; **C** = captioning; **E** = editing prompt reconstruction; **G** = generating prompt reconstruction; **L** = labeling; **P** = preference; **R** = rating.

Table 5: Pre-2015 publications.

Name	Citations	License	Elements	Eval.
ImageNet [1]	66,633	FNCR	Img + Lbl	L
UCF101 [2]	6,460	CC0-1.0	Vid + Act	L
MS COCO [3]	46,361	CC BY 4.0	Img + Seg + Cap	B + C

Table 6: 2015 publications.

Name	Citations	License	Elements	Eval.
Flickr30k Entities [4]	2,226	FNCR	Img + Cap + BB + Rel	B + C
FlyingThings3D [5]	2,758	FNCR	Img pairs + 3D	A
LSUN [6]	2,436	FNCR + conditions	Img + Lbl	L

Table 7: 2016 publications.

Name	Citations	License	Elements	Eval.
Cityscapes [7]	12,189	FNCR	Img + Seg + Lbl	L
Visual Genome [8]	6,018	CC BY 4.0*	Img + BB + Rel + QA	A
VQA v2.0 [9]	3,558	CC BY 4.0	Img pairs + QA	A

Table 8: 2017 publications.

Name	Citations	License	Elements	Eval.
Fashion-MNIST [10]	9,435	MIT	Img + Lbl	L
Kinetics Human Action Video Dataset [11]	3,981	CC BY 4.0*	Vid + Lbl	L
iNaturalist [12]	1,763	CC BY-NC	Img + BB + Lbl	L + B

Table 9: 2018 publications.

Name	Citations	License	Elements	Eval.
Places [13]	4,319	FNCR	Img + Lbl	L
Conceptual Captions [14]	2,643	FAP	Img + Cap	C
Open Images V4 [15]	1,441	CC BY 2.0 (Img) / 4.0 (ann.)	Img + BB + Lbl + Rel	L + B

Table 10: 2019 publications.

Name	Citations	License	Elements	Eval.
GQA [16]	2,402	CC BY 4.0	Img + QA	A
MVTec AD [17]	1,549	CC BY-NC-SA 4.0	Img + Seg	L
LVIS [18]	1,492	CC BY 4.0	Img + Seg + Lbl	B

Table 11: 2020 publications.

Name	Citations	License	Elements	Eval.
DocVQA [19]	930	Unk.	Img + QA	A
DFDC [20]	597	MRL?	Vid + Lbl	L
TextCaps [21]	469	CC BY 4.0	Img + Cap	C

Table 12: 2021 publications.

Name	Citations	License	Elements	Eval.
LAION-400M [22]	1,486	CC BY 4.0 (meta.); Img: OC	Img + Lbl + Sc	C
FairFace [23]	616	CC BY 4.0	Img + Lbl	L
HDTF [24]	413	GPL-3.0	Vid + Aud + Txt	C

Table 13: 2022 publications.

Name	Citations	License	Elements	Eval.
LAION-5B [25]	4,059	CC BY 4.0 (meta.); Img: OC	Img + Cap + Sc	C
DiffusionDB [26]	349	CC0-1.0 (data) / MIT (code)	Img + Pr + Sc	G
TAD66K [27]	111	Apache-2.0	Img + Sc	R

Table 14: 2023 publications.

Name	Citations	License	Elements	Eval.
MagicBrush [28]	391	CC BY-SA 4.0	Img pairs + edit Pr	E
Pick-a-Pic [29]	559	Unk.	Img + Pr + Pref	P
InternVid [30]	411	CC BY-NC-SA 4.0	Vid + Cap	C

Table 15: 2024 publications.

Name	Citations	License	Elements	Eval.
OpenVid-1M [31]	149	Unk.	Vid + Pr + Cap	C + G
Visual CoT [32]	148	CC BY-SA 4.0	Img + BB + QA	A + B
HQ-Edit [33]	104	CC BY-NC-4.0	Img pairs + edit Pr	E

Table 16: 2025 publications.

Name	Citations	License	Elements	Eval.
BLIP3o-60k [34]	91	Apache-2.0	Img + Pr	G
ImgEdit [35]	38	Apache-2.0	Img pairs + edit Pr	E
ShareGPT-4o-Image [36]	27	Apache-2.0	Img + Pr	G + E

In the dataset summary tables, when images are grouped or classified into categories, we count this as having ***labels***. Labels are category tags that describe what the image or video shows. For example, a picture of a dog in **ImageNet** has labels such as `dog`, `mammal`, `animal`, while a picture of a woman laughing in **UCF101** is labeled as `smiling`. ***Captions*** go a step further by describing what is happening or what can be seen in the image or video, as seen in Figure 14 for an image captioned "`pop artist performs at the festival in a city`" in the **Conceptual Captions** dataset.***Prompts*** are pieces of text used to generate or edit images in datasets built around generative models. Examples can be seen in Figures 26 or 28 for the **DiffusionDB** or **MagicBrush** datasets respectively.

Bounding boxes pinpoint specific object instances within an image by drawing rectangles around them (using coordinates such as $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$). See Figures 4 or 8 from **Flickr30K** or **Visual Genome** respectively as examples. ***Segmentation*** goes a step further by outlining the exact shape of each object, assigning every pixel to a class or instance. Semantic segmentation labels each pixel by class, while instance segmentation gives each object its own mask. See Figures 3 or 7 from **MS COCO** or **Cityscapes** respectively for examples.

Finally, **Pick-a-Pic** captures human ***preferences*** for images generated by text-to-image diffusion models. Images in the dataset come in pairs, with one of the two image being preferred. Figure 29 shows examples where the chosen image is highlighted and the rejected one is darkened. The term ***scores*** refers to any numerical metadata. For example, an aesthetic score as in **TAD66K** or an NSFW score as in **LAION-5B**. ***QA-pairs*** (question–answer pairs) are exactly what they sound like: a question about the image and its corresponding answer, such as "`What type of fruit in the image is round?`" and "`apple`", as shown in Figure 16 for the **GQA** dataset.

Relationships refer to datasets that explicitly describe how objects interact or relate to one another, rather than just listing them. Only two datasets meet this criterion: **Flickr30K Entities** and **Visual Genome**. The former connects phrases in captions that refer to the same entity (like "`a boy`" and "`the child`") and links them to specific bounding boxes in the image, as shown in Figure 4. The latter represents scenes as graphs of objects and their relationships e.g., "`man` *`riding`* `horse`", as shown in Figure 8. Finally, when a dataset contains ***3D data***, it includes information about the geometry of the scene beyond standard RGB images. This only applies to **FlyingThings3D**, which contains stereo image pairs (left and right views rendered from two virtual cameras) along with detailed geometric information such as depth (disparity) maps, optical flow, segmentation masks, material indices, motion boundaries, and camera calibration data. Together, these describe the full 3D structure and motion of the scene, as shown in Figure 5.

Regarding the licenses,

6.3 Benchmark Tasks and Metrics

6.3.1 L: Labeling

This benchmark takes one image and a list of possible labels. The model returns exactly one label describing the image. For video datasets, several frames are combined into a contact sheet and the model predicts the action or video label.

The smallest context windows among the listed models appear to be those of `glm-4.5v-thinking` and `glm-4.1v-thinking`, both at 64k tokens. Depending on whether one uses the official serving configuration rather than the Hugging Face architectural config, `dots.vlm1` may also be treated as a 65,536-token model.

Among the classification datasets considered in this study, the largest label spaces correspond to

ImageNet-1K, iNaturalist, and Open Images V4. ImageNet-1K contains 1,000 classes, iNaturalist contains 5,089 species categories, and Open Images V4 provides approximately 20,000 image-level concepts. Since all evaluated models support context windows of at least 64k input tokens, the complete label set can be provided as part of the prompt for ImageNet-1K without difficulty.

For iNaturalist, only the species-level labels will be included in the prompt. This reduces the required prompt length to approximately 20k–30k tokens while preserving the standard fine-grained classification task. Higher levels of the taxonomic hierarchy (e.g., genus, family, order, and class) will not be provided as candidate labels.

For Open Images V4, the full image-level vocabulary is too large to be comfortably included within a 64k-token context window. Therefore, only the subset of images associated with bounding-box annotations will be used. This restricts the candidate label set to the 600 object categories employed by the detection benchmark, allowing all possible labels to be supplied in the prompt while maintaining a closed-set classification setting.

6.3.2 A: Answering Visual Multiple-choice Question

This benchmark takes an image, a natural-language question about its contents, and a set of possible answers. The model selects one answer, which is normalized and compared with the accepted reference answer.

All VQA is multiple choice. For some dataset, distractor choices had to be manually created. For others, even the correct choices had to be created.

The distractor choices must be created for **Visual Genome**, **VQA v2.0**, **GQA**, **DocVQA**, and **Visual CoT**. These datasets already provide each image, question, and accepted answer, so the accepted answer can be used as the correct choice; however, their dataset adapters do not provide a set of incorrect candidate answers. Consequently, plausible but incorrect distractors must be added to convert these open-answer questions into multiple-choice questions.

FlyingThings3D requires more extensive construction because it is not originally a question–answer dataset. It provides rendered stereo scenes and geometric annotations rather than natural-language questions with accepted answers. For this dataset, the question and correct textual choice must first be derived from the available scene information; the incorrect choices must then also be created. Thus, only the distractors need to be added for the five existing QA datasets, whereas both the benchmark question/answer and its distractors need to be constructed for FlyingThings3D.

Visual Genome has bouding boxes, but these are very inconsitent because they are human anotatesd without a standard list of labels, so we may get ‘pet dog‘ somewhere, but ‘small dog‘ elsewhere. That is why we will not measure it in terms of boudning boxes. Moreover, there are over 30k different types of annotated objects, which would not fit into our maximum of 64k labels.

6.3.3 B: Bounding Box Object Detection

This benchmark takes an image and a list of allowed object labels. The model returns every detected object as a label and normalized bounding box. Predictions are compared with annotated boxes using label agreement and intersection over union (IoU).

6.3.4 C: Captioning

This benchmark takes an image, or representative frames from a video, without answer choices. The model generates one concise natural-language description. It is compared with one or more reference captions using BLEU.

6.3.5 E: Editing Prompt Reconstruction

This benchmark takes a side-by-side pair containing the original image and its edited version, plus four candidate editing prompts. The model selects the prompt that best explains the transformation from the first image to the second.

6.3.6 G: Generating Prompt Reconstruction

This benchmark takes an image generated from text and four candidate generation prompts. The model selects the prompt that most likely produced the image, returning either its choice letter or the exact prompt.

6.3.7 P: Preference

This benchmark takes two images shown side by side. The model chooses which image is preferred by returning Image A or Image B. The choice is compared with the recorded human preference.

6.3.8 R: Rating

This benchmark takes one image and asks the model to assess its overall aesthetic quality. The model returns one integer from 1 to 10. Evaluation records exact agreement and the absolute difference from the reference rating.

7 Design and Implementation

7.1 Experimental Infrastructure and Deployment

7.1.1 Execution Plan for Open-Weight Models

The open-weight evaluation will use a single NVIDIA B300 GPU on RunPod. A model will be included only if its highest-precision checkpoint and the additional memory required for inference fit on that GPU. Models that require more than one B300 will not be evaluated: hosted inference, multi-GPU execution, and multi-node deployment are outside the scope of this study. The oversized models remain listed in Table 17 to document why they were excluded from the experimental results.

A single NVIDIA B300 GPU provides 288 GB of HBM3e memory [41]. The VRAM estimates in Table 17 refer only to the memory required to store the model weights. They do not include inference overhead, KV cache, image-token memory, batching overhead, CUDA graphs, or framework-specific runtime memory. Therefore, models whose weights are close to 288 GB are treated as borderline even if the raw weight estimate is technically below 288 GB.

For a model with N parameters and b bits per parameter, the weight-only VRAM requirement is estimated as

$$\mathrm{VRAM}_{\mathrm{weights}} \approx N \cdot \frac{b}{8}.$$

For mixture-of-experts models, the estimate uses the total number of parameters, not only the number of active parameters, since the full set of expert weights must normally be available during inference.

Table 17: Execution plan for open-weight models under a single-B300 RunPod setup.

Model	Native precision / size	Weight-only VRAM	Execution plan
`dots.ocr`	~ 2.9B, 16-bit	~ 5.8 GB	RunPod, single B300
`glm-4.1v-thinking`	~ 9B, 16-bit	~ 18 GB	RunPod, single B300
`gemma-4-26b-a4b`	~ 26B, 16-bit	~ 52 GB	RunPod, single B300
`qwen3.6-27b`	~ 27B, 16-bit	~ 54 GB	RunPod, single B300
`gemma-4-31b`	~ 31B, 16-bit	~ 62 GB	RunPod, single B300
`skywork-r1v2-38b`	~ 38B, 16-bit	~ 76 GB	RunPod, single B300
`skywork-r1v3-38b`	~ 38B, 16-bit	~ 76 GB	RunPod, single B300
`mistral-small-4`	~ 119B, native FP8	~ 119 GB	RunPod, single B300
`glm-4.5v-thinking`	~ 106B, 16-bit	~ 212 GB	RunPod, single B300
`mistral-medium-3.5`	~ 128B, 16-bit	~ 256 GB	RunPod, single B300,
`mimo-v2.5`	~ 310B, native FP8	~ 310 GB	Excluded: exceeds one
`qwen3-vl-235b-a22b`	~ 235B, 16-bit	~ 470 GB	Excluded: exceeds one
`kimi-k2.5-thinking`	~ 1T, native mixed INT4	~ 550–600 GB	Excluded: exceeds one
`kimi-k2.6`	~ 1T, native mixed INT4/NVFP4	~ 600–714 GB	Excluded: exceeds one
`dots.vlm1`	~ 672B, FP8 deployment	~ 672 GB	Excluded: exceeds one
`qwen3.5-397b-a17b`	~ 397B, 16-bit	~ 794 GB	Excluded: exceeds one
`mimo-v2.5-pro`	~ 1.02T, native FP8	~ 1020 GB	Excluded: exceeds one

Under this plan, every included open-weight model is evaluated with the same datasets, prompt templates, image preprocessing, answer extraction code, scoring metrics, logging format, and single-B300 RunPod environment. A raw weight estimate below 288 GB does not guarantee inclusion, because inference also requires memory for the KV cache, image tokens, temporary activations, and framework overhead. Therefore, a borderline model such as `mistral-medium-3.5` will be included only if the complete inference pipeline runs successfully on one B300 without reducing the checkpoint below its stated native precision. If it does not, it will be reported as excluded rather than moved to a hosted or multi-GPU backend.

For this reason, each benchmark result should include at least the following metadata:

model identifier	exact checkpoint name,
backend	self-managed inference on RunPod,
precision	BF16, FP8, INT4, or checkpoint-declared,
hardware	single NVIDIA B300,
date tested	date on which the result was obtained.

8 Finetuning

8.1 Purpose and Model Selection

The finetuning experiment measures whether parameter-efficient adaptation to a balanced mixture of visual tasks improves performance across the eight benchmark types defined in Section 6.3, rather than only on a single dataset. The selected checkpoint is `LLaVA-OneVision-Qwen2-7B`[1], referred to below as LLaVA-OneVision 7B. This model was selected because it is supported directly by the Hugging Face `transformers` library, accepts image–text conversations, and is sufficiently small for both inference and parameter-efficient finetuning on one GPU.

Two notebooks implement a matched before-and-after experiment. The baseline notebook evaluates the unchanged checkpoint, while the finetuning notebook constructs the training data, trains one adapter, and evaluates the adapted model. Both notebooks use the same model checkpoint, benchmark classes, evaluation splits, number of evaluation examples, generation limits, and quantization setting. Consequently, the main experimental difference is the presence or absence of the trained adapter.

8.2 Representative Training Tasks

One representative dataset is selected for each benchmark type. The training mixture is therefore broad enough to expose the model to all required answer formats while remaining balanced across task families.

Type	Task	Representative dataset
L	Labeling	Fashion-MNIST
A	Visual question answering	DocVQA
B	Bounding-box detection	Open Images V4
C	Captioning	MS COCO captions
E	Editing prompt reconstruction	MagicBrush
G	Generation prompt reconstruction	DiffusionDB
P	Image preference	Pick-a-Pic
R	Aesthetic rating	TAD66K

Table 18: Representative datasets used in the eight-type finetuning mixture.

By default, 980 training examples and 200 validation examples are exported for each type. This produces a 7,840-example training mixture and a 1,600-example validation mixture. Equal per-type counts prevent a large source dataset from dominating the optimization objective. After export, the eight manifests are merged and deterministically shuffled. The training count is limited by the configured DocVQA mirror, which exposes 1,286 usable validation images: after reserving 100 for evaluation and 200 for finetuning validation, 980 are used for training, leaving a small safety margin.

8.3 Conversion to Supervised Conversations

The same benchmark implementation is used to construct both evaluation inputs and finetuning examples. For every selected row, the exporter:

1. loads and normalizes the visual input;
2. calls the benchmark's prompt-construction method;
3. obtains the accepted answer from the dataset annotations;

[1] `https://huggingface.co/llava-hf/llava-onevision-qwen2-7b-ov-hf`

4. writes the image path, prompt, answer, and system instruction to a JSONL manifest.

Each exported record has the conceptual form

$$(\text{image},\ \text{task instruction},\ \text{target answer}).$$

During training, this record is rendered with the LLaVA-OneVision chat template. The user message contains the image and task instruction, and the assistant message contains the accepted answer. Tokens belonging to the system and user messages are masked from the loss. Therefore, the optimization loss is computed only on the assistant answer, rather than teaching the model to reproduce the entire prompt.

The target format depends on the task. Labeling examples contain one class name; visual-question-answering examples contain the accepted answer; detection examples contain one or more labels and normalized $[x, y, \text{width}, \text{height}]$ boxes; captioning examples contain a reference caption; preference examples contain `Image A` or `Image B`; and rating examples contain the reference score.

8.4 Manually Constructed Questions and Distractors

Some benchmarks require information that is not present in the source dataset. This occurs mainly when an originally open-answer or prompt-reconstruction task is converted into multiple choice. It is important to distinguish the *supervised answer* from the *manually constructed choices.*

For Type A datasets such as DocVQA, the source dataset already supplies the image, question, and accepted answer. Manual curation supplies plausible incorrect choices for selected rows; it does not replace the source answer. When curated choices are available, the prompt lists those choices and the target remains the accepted dataset answer. When no curated choices are available, the exporter uses the native open-answer question and answer.

For Type E, MagicBrush supplies a source image, an edited image, and the edit instruction that produced the transformation. The correct edit instruction is the supervised target. Manually curated instructions, dataset-derived instructions, or generic fallback instructions are used only as distractors in the multiple-choice prompt.

Type G follows the same principle. DiffusionDB supplies the generated image and its original text prompt. That original prompt is the target answer, while the other candidate prompts are distractors. Type P does not require manually written prompts: Pick-a-Pic supplies the compared images and the recorded human preference, so the target is simply `Image A` or `Image B`.

Thus, the model is never trained to generate the manually authored distractors. It is trained to predict the source dataset's correct answer in the context of the benchmark prompt. Manual curation controls the difficulty and validity of the question presented to the model.

The curated choice files cover a limited number of rows rather than every possible training example. The finetuning exporter therefore supports a `skip-examples` parameter. The first 100 examples are reserved for the matched evaluation, and training export begins after this reserved prefix. As a result, manually curated rows at the beginning of a dataset are primarily used for evaluation, while later finetuning rows use native open-answer supervision or automatically constructed choices. This reduces direct memorization of the small manual prompt bank.

8.5 QLoRA Training Procedure

The experiment trains one balanced adapter shared by all eight task types. The base checkpoint is loaded in 4-bit NormalFloat quantization and adapted with QLoRA. Low-rank adapters are inserted into the model's linear layers with rank 32, scaling factor 64, and dropout 0.05. The vision tower is frozen by default so that optimization concentrates on the language model and multimodal projection components. The default run uses two epochs, a learning rate of 10^{-4}, a physical batch size of one, and gradient accumulation over 16 steps.

Training uses BF16 computation, gradient checkpointing, a cosine learning-rate schedule, and validation at the end of each epoch. Checkpoints are selected by minimum validation loss. The final output contains the PEFT adapter and the processor configuration; it does not duplicate the complete base model. Evaluation therefore reloads the original LLaVA-OneVision checkpoint and then attaches the trained adapter.

8.6 Baseline and Adapted Evaluation

The unchanged model and adapted model are each evaluated on 100 examples from every representative benchmark. Results are written separately for the baseline and finetuned runs, using the repository's standard benchmark runner and scoring code. The experiment therefore produces eight baseline reports and eight adapted reports. Improvement is assessed per benchmark and per task type rather than only through one aggregate score, since a shared adapter may improve some output formats while degrading others.

Where a dataset provides a dedicated split, training and evaluation use different splits. Fashion-MNIST trains on `train` and evaluates on `test`; Open Images detection trains on `train` and evaluates on `validation`; and MS COCO captioning trains on `train` and evaluates on `test`. Other repository adapters expose only one practical split. For these datasets, the exporter reserves the initial evaluation-sized slice before creating the training and validation manifests.

8.7 Validity Limitations

Prefix reservation is weaker than an official held-out split. Some streaming datasets may change row order or sampling behavior when a different number of examples is requested. Therefore, DocVQA and the train-only MagicBrush, DiffusionDB, Pick-a-Pic, and TAD66K comparisons must be interpreted as exploratory unless stable row identifiers are recorded and checked for disjointness. The strongest before-and-after evidence comes from datasets with officially separate training and evaluation splits.

The experiment also measures adaptation to both visual reasoning and output formatting. For example, an improvement on detection may result partly from learning the required normalized-box syntax, while an improvement on multiple-choice reconstruction may result partly from learning to return a choice letter or exact option. Per-task error analysis is therefore required to distinguish improved visual understanding from improved compliance with the benchmark response format.

9 Tests, Evaluation, and Results

10 Conclusions and Future Work

11 References

References

[1] J. Deng et al., "ImageNet: A Large-Scale Hierarchical Image Database," CVPR, 2009. https://doi.org/10.1109/CVPR.2009.5206848.

[2] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A Dataset of 101 Human Actions Classes From Videos in the Wild," 2012. https://arxiv.org/abs/1212.0402.

[3] T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context," ECCV, 2014. https://arxiv.org/abs/1405.0312.

[4] B. A. Plummer et al., "Flickr30k Entities: Collecting Region-to-Phrase Correspondences for Richer Image-to-Sentence Models," ICCV, 2015. https://arxiv.org/abs/1505.04870.

[5] N. Mayer et al., "A Large Dataset to Train Convolutional Networks for Disparity, Optical Flow, and Scene Flow Estimation," CVPR, 2016. https://arxiv.org/abs/1512.02134.

[6] F. Yu et al., "LSUN: Construction of a Large-scale Image Dataset using Deep Learning with Humans in the Loop," 2015. https://arxiv.org/abs/1506.03365.

[7] M. Cordts et al., "The Cityscapes Dataset for Semantic Urban Scene Understanding," CVPR, 2016. https://arxiv.org/abs/1604.01685.

[8] R. Krishna et al., "Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations," IJCV, 2017. https://arxiv.org/abs/1602.07332.

[9] Y. Goyal et al., "Making the V in VQA Matter," CVPR, 2017. https://arxiv.org/abs/1612.00837.

[10] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms," 2017. https://arxiv.org/abs/1708.07747.

[11] W. Kay et al., "The Kinetics Human Action Video Dataset," 2017. https://arxiv.org/abs/1705.06950.

[12] G. Van Horn et al., "The iNaturalist Species Classification and Detection Dataset," CVPR, 2018. https://arxiv.org/abs/1707.06642.

[13] B. Zhou et al., "Places: A 10 Million Image Database for Scene Recognition," TPAMI, 2017. https://arxiv.org/abs/1610.02055.

[14] P. Sharma et al., "Conceptual Captions: A Cleaned, Hypernymed, Image Alt-text Dataset for Automatic Image Captioning," ACL, 2018. https://aclanthology.org/P18-1238/.

[15] A. Kuznetsova et al., "The Open Images Dataset V4," IJCV, 2020. https://arxiv.org/abs/1811.00982.

[16] D. A. Hudson and C. D. Manning, "GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering," CVPR, 2019. https://arxiv.org/abs/1902.09506.

[17] P. Bergmann et al., "MVTec AD—A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection," CVPR, 2019. https://arxiv.org/abs/1904.09665.

[18] A. Gupta, P. Dollar, and R. Girshick, "LVIS: A Dataset for Large Vocabulary Instance Segmentation," CVPR, 2019. https://arxiv.org/abs/1908.03195.

[19] M. Mathew, D. Karatzas, and C. V. Jawahar, "DocVQA: A Dataset for VQA on Document Images," WACV, 2021. https://arxiv.org/abs/2007.00398.

[20] B. Dolhansky et al., "The DeepFake Detection Challenge (DFDC) Dataset," 2020. https://arxiv.org/abs/2006.07397.

[21] O. Sidorov et al., "TextCaps: a Dataset for Image Captioning with Reading Comprehension," ECCV, 2020. https://arxiv.org/abs/2003.12462.

[22] C. Schuhmann et al., "LAION-400M: Open Dataset of CLIP-Filtered 400 Million Image-Text Pairs," 2021. https://arxiv.org/abs/2111.02114.

[23] K. Karkkainen and J. Joo, "FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age," WACV, 2021. https://arxiv.org/abs/1908.04913.

[24] Z. Zhang et al., "Flow-guided One-shot Talking Face Generation with a High-Definition Talking Face Dataset," CVPR, 2021. https://github.com/MRzzm/HDTF.

[25] C. Schuhmann et al., "LAION-5B: An Open Large-Scale Dataset for Training Next Generation Image-Text Models," NeurIPS Datasets and Benchmarks, 2022. https://arxiv.org/abs/2210.08402.

[26] Z. J. Wang et al., "DiffusionDB: A Large-scale Prompt Gallery Dataset for Text-to-Image Generative Models," ACL, 2023. https://arxiv.org/abs/2210.14896.

[27] S. He et al., "Theme-Aware Visual Attribute Reasoning for Image Aesthetics Assessment," ACM MM, 2022. https://github.com/woshidandan/TANet.

[28] K. Zhang et al., "MagicBrush: A Manually Annotated Dataset for Instruction-Guided Image Editing," NeurIPS Datasets and Benchmarks, 2023. https://arxiv.org/abs/2306.10012.

[29] Y. Kirstain et al., "Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation," NeurIPS, 2023. https://arxiv.org/abs/2305.01569.

[30] Y. Wang et al., "InternVid: A Large-scale Video-Text Dataset for Multimodal Understanding and Generation," 2023. https://arxiv.org/abs/2307.06942.

[31] K. Nan et al., "OpenVid-1M: A Large-Scale High-Quality Dataset for Text-to-video Generation," 2024. https://arxiv.org/abs/2407.02371.

[32] H. Shao et al., "Visual CoT: Advancing Multi-Modal Language Models with a Comprehensive Dataset and Benchmark for Chain-of-Thought Reasoning," 2024. https://arxiv.org/abs/2403.16999.

[33] M. Hui et al., "HQ-Edit: A High-Quality Dataset for Instruction-based Image Editing," 2024. https://arxiv.org/abs/2404.09990.

[34] J. Chen et al., "BLIP3-o: A Family of Fully Open Unified Multimodal Models—Architecture, Training and Dataset," 2025. https://arxiv.org/abs/2505.09568.

[35] Y. Ye et al., "ImgEdit: A Unified Image Editing Dataset and Benchmark," 2025. https://arxiv.org/abs/2505.20275.

[36] J. Chen et al., "ShareGPT-4o-Image: Aligning Multimodal Models with GPT-4o-Level Image Generation," 2025. https://arxiv.org/abs/2506.18095.

[37] R. Kinney et al., "The Semantic Scholar Open Data Platform," 2023. https://arxiv.org/abs/2301.10140.

[38] X. Yue et al., "MMMU-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark," 2024. https://arxiv.org/abs/2409.02813.

[39] W.-L. Chiang et al., "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference," ICML, 2024. https://arxiv.org/abs/2403.04132.

[40] LM Arena, "Vision Arena Leaderboard." https://lmarena.ai/leaderboard/vision. Accessed June 10, 2026.

[41] NVIDIA, "NVIDIA B300 Tensor Core GPU." https://www.nvidia.com/en-us/data-center/b300/. Accessed June 10, 2026.

A Detailed Dataset Descriptions

A.1 Dataset Catalogue

ImageNet [1] is a large-scale database of human-annotated photographs organized into a hierarchy of visual concepts. Each category, or "synset," groups together images of the same concept, such as "husky", and is linked to related categories within a larger hierarchy, such as "animal". At the time of publication, ImageNet contained 12 major groups, covering 5,247 categories and about 3.2 million images.

Figure 2: Sample frames from UCF101 showing diverse action classes such as *Apply Eye Makeup*, *Playing Dhol*, and *Baby Crawling*.

UCF101 (2012) [2]. The University of Central Florida (UCF) compiled a total of 13,320 video clips amounting to about 27 hours of footage. Each clip was then labeled as one of 101 action classes, 51 of which preserved audio. The videos are stored at a resolution of 320×240 pixels and 25 frames per second. On average, clips are about 7 seconds long.

Figure 3: Examples from the MS COCO dataset showing object segmentations for categories such as *cow*, *train*, and *car*

Microsoft COCO (2014) [3]. The Microsoft COCO, abbreviated MS COCO, dataset contains photographs of everyday environments filled with multiple objects in their natural context. Each object instance is carefully annotated with detailed segmentations, allowing precise localization within the image as shown in Figure 3.

MS COCO includes 91 common object categories that are easily recognizable, with over 2.5 million labeled instances across 328,000 images.

Figure 4: Example image with captions; phrases in the same color are coreferent and linked to their corresponding bounding boxes.

Flickr30k Entities [4] extends the Flickr30k dataset by providing detailed links between image regions and textual phrases. With 31,783 images and 158,915 captions, the dataset connects mentions that refer to the same entity across captions describing the same image,

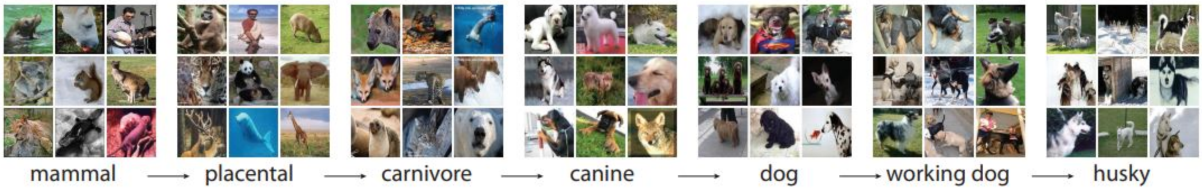

Figure 1: Example of ImageNet hierarchical organization, showing a path from *mammal* to *husky*.

forming about 244,000 coreference chains, and associates these with roughly 276,000 manually drawn bounding boxes. An example of the final annotated data is shown in Figure 4, where color-coded phrases correspond to specific image regions. These annotations enable the new task of *phrase localization*: given an image and a caption, predict the region that corresponds to a particular entity mention.

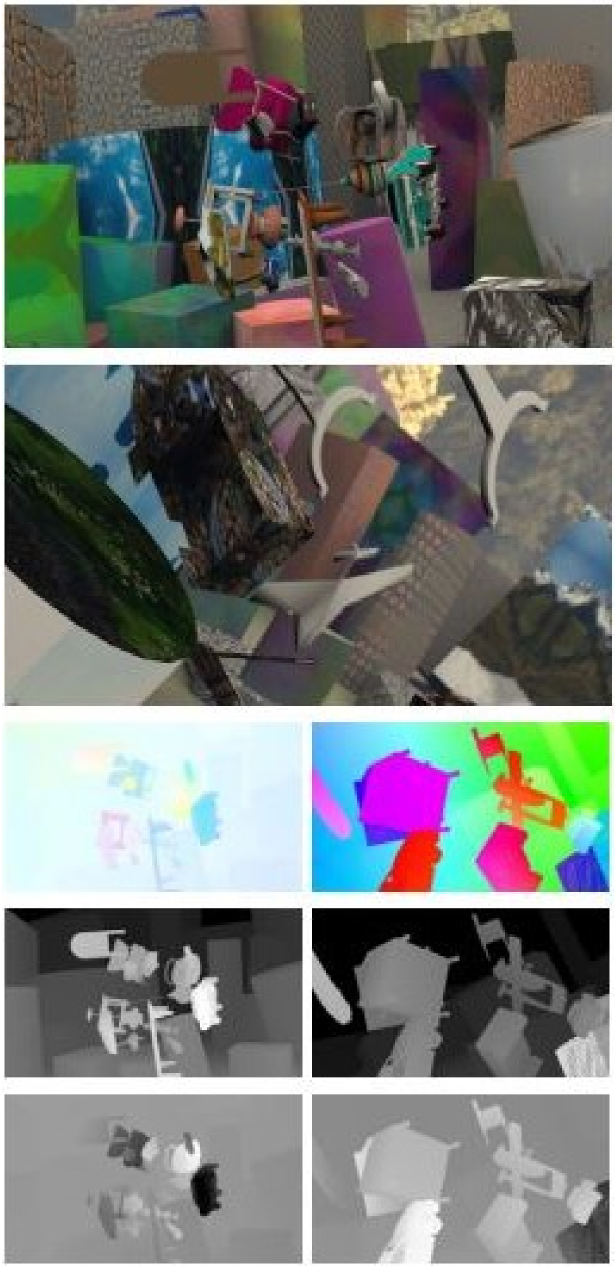

Figure 5: Example scenes from the *FlyingThings3D* dataset. The third row shows optical flow images, the fourth row shows disparity images, and the fifth row shows disparity change images (normalized for display).

FlyingThings3D [5] consists of roughly 25,000 stereo frames rendered using *Blender*, each dis-

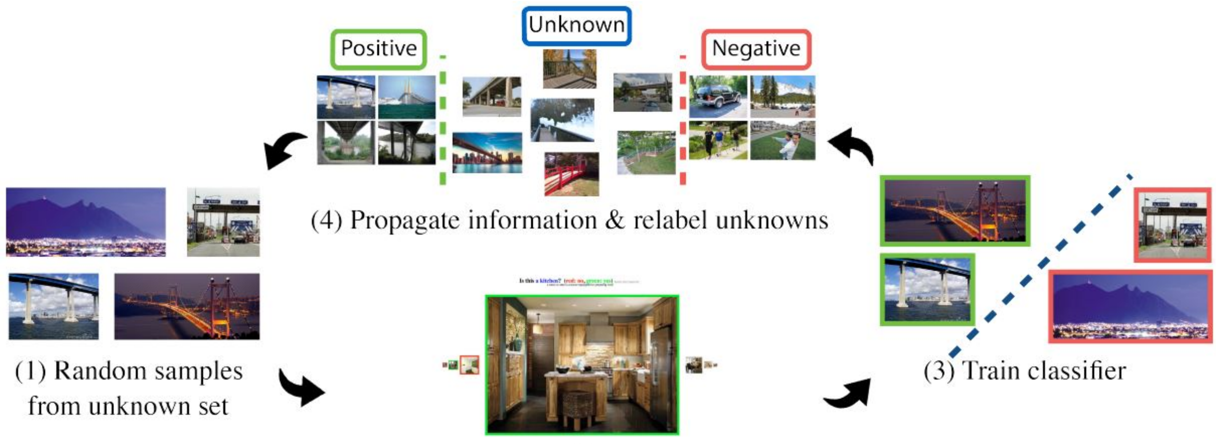

Figure 6: Example images and scene categories from the *LSUN* dataset.

Figure 7: Example scenes from the *Cityscapes* dataset. From left to right: fine annotations for training/validation (3,475 images), coarse annotations for training (20,000 images), and fine annotations for testing (1,525 images). Each image shows dense semantic and instance-level labeling of complex urban street scenes.

playing a random arrangement of textured 3D objects. Each frame provides left and right RGB images, together with corresponding maps for optical flow, disparity, and disparity change, as well as object segmentation masks, material indices, and motion boundaries as shown in Figure 5. Complete camera calibration data and 3D geometry information are also provided.

LSUN [6] is a very large dataset consisting of about 69 million images belonging to 30 categories. Only a small portion of the images were labeled by humans, the rest being labeled by a trained classifier (e.g., for the *train* category, less than 1/60 of images were labeled by people), achieving 90% precision. Training popular convolutional networks on this larger, if noisier, data leads to noticeable gains; for example, fine-tuning with LSUN reduced a reported classification error from 28.6% to 22.2% on a related evaluation.

Cityscapes [7] contains stereo video sequences recorded across 50 different cities, primarily in Germany. From these recordings, 5,000 images received high-quality fine annotations, while an additional 20,000 were annotated coarsely (less boundary precision) with a 98% accuracy after excluding ambiguous (void) regions. Figure 7 illustrates examples of both fine and coarse annotations used for training, validation, and testing. The dataset is split into 2,975 training, 500 validation, and 1,525 test images, ensuring balance across geography, population size, and season. In total, 30 semantic classes (19 for evaluation) are grouped into eight high-level categories: flat, construction, nature, vehicle, sky, object, human, and void.

Visual Genome [8] human-annotated imagea describe not only *what* objects are present, but also *how* they interact. The dataset contains over 100,000 real-world images, each annotated with an average of 21 objects, 18 attributes, and 18 pairwise relationships, as exemplified in Figure 8. Beyond scene graphs, Visual Genome includes over 1.7 million question–answer pairs designed to test visual reasoning and comprehension.

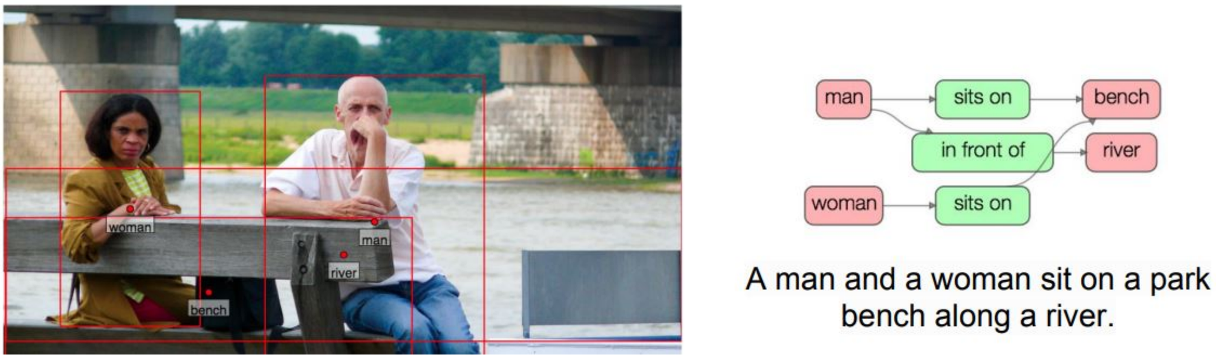

Figure 8: Example image and scene graph from the **Visual Genome** dataset. The left side shows labeled objects with bounding boxes, while the right side illustrates the relationships between these objects in a graph.

Figure 9: Examples from the balanced **VQA v2.0** dataset. Each question is paired with images that are visually similar yet lead to different answers, encouraging models to use visual evidence rather than language priors.

VQA v2.0 [9]. For each image in the Visual Question Answering (VQA) dataset, annotators tried to find a complementary image where the question still makes sense, but has a different answer. These image pairs with question-answers result in VQA v2.0, as shown in Figure 9. In total, they collected approximately 195K complementary images for train, 93K complementary images for val, and 191K complementary images for test set.

Benchmarking shows that state-of-the-art models trained on the original data perform significantly worse on the balanced set; clear evidence that they had exploited language regularities. Training on the balanced data improves results but the task remains challenging.

Fashion-MNIST [10] consists of 70,000 grayscale images of size 28 × 28, evenly distributed across 10 fashion categories such as T-shirts, trousers, coats, and shoes. The training set contains 60,000 images, and the test set includes 10,000 images. As such, the dataset is similar in structure to the MNIST digits dataset, from where it gets its name. Fashion-MNIST poses a significantly harder classification challenge: typical models that achieve over 99.7% accuracy on digits reach only around 89–90% on Fashion-MNIST.

The **Kinetics Human Action Video Dataset** consists of 400 action classes with approximately 400–1150 ten-second clips per class, each extracted from a distinct YouTube video (for a total of ∼306K clips). Actions span three major categories: *person-only actions* (e.g., laughing, brushing hair), *person–object interactions* (e.g., playing instruments), and *person–person interactions* (e.g., shaking hands). Clips retain audio but are designed for visual classification, enabling use in multimodal learning tasks. The dataset is divided into three subsets with 250–1000 training clips per class, 50 validation clips per class, and 100 test clips per class.

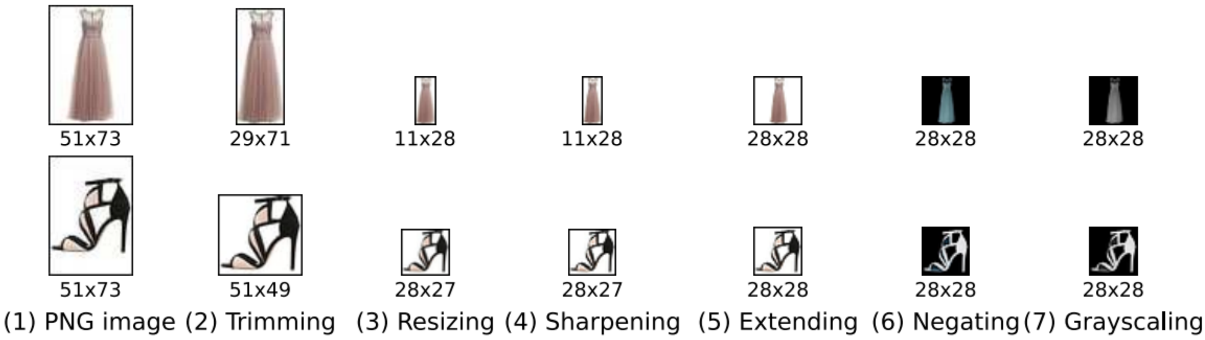

Figure 10: The conversion pipeline used to generate the *Fashion-MNIST* dataset. Each row shows an example from a product category (dress, sandal), with transformations applied sequentially until the final (rightmost) image that will be part of the dataset is created.

Figure 11: Example frames from the "brushing hair" class in the **Kinetics Human Action Video Dataset**, illustrating variation in appearance, lighting, and camera viewpoint. Each clip corresponds to a unique YouTube video.

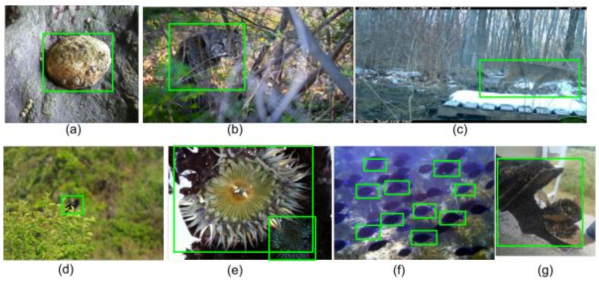

Figure 12: Sample bounding-box annotations for the **iNaturalist** dataset. Annotators drew boxes for the *super-class* (e.g., "box all birds"), up to ten instances per image; boxes could overlap, cover only visible parts when occluded, and a single box could enclose physically connected instances.

iNaturalist (2017) [12], at time of publication, contained 859,000 images (579,184 train, 95,986 val, and 182,707 test images) spanning 5,089 plant and animal species. Labels reflect the consensus of informed contributors on the iNaturalist platform, and annotators drew boxes around the plant/animals that appear on the image. On classes with the most training examples, the best classifiers at the time could only reach 67% accuracy.

The **Places** database [13] comprises 10 million images labeled with 434 semantic scene categories (e.g., "messy kitchen", "misty coast"). Designed to complement object-centric datasets like ImageNet, it provides exhaustive coverage of environments such as *beaches*, *corridors*, *kitchens*, and *forests*, enabling models to understand not just objects, but the broader spatial and functional context in which they appear. An unusual and creative element of the collection process was the use of 696 English adjectives to diversify visual queries — combinations such as "messy kitchen," "romantic bedroom," and "misty coast."

Figure 13: Example scenes from the *Places* dataset, illustrating variations across categories such as bedrooms, kitchens, forests, and coasts. Each scene combines structural and stylistic diversity captured through adjective-based search expansions.

Figure 14: Examples from the **Conceptual Captions** dataset. Original web `alt`-texts (top) are automatically cleaned and hypernymed into concise captions (bottom).

The **Conceptual Captions** dataset [14] contains about 3.3 million image–caption pairs automatically harvested from web `alt`-texts, making it over ten times larger than **MS COCO**. The final split includes 3.3 million training examples, 28,000 validation, and 22,500 test pairs, with captions averaging ten words in length.

The dataset was created through a multi-stage automatic pipeline that processed billions of webpages. The first stage filtered out non-JPEG, small, distorted, or offensive images. In the second stage, noisy or incomplete `alt`-texts—such as keyword spam or hashtag lists—were discarded using part-of-speech, sentiment, and profanity analysis. Next, candidate captions were retained only if there was lexical overlap between concepts detected in the image (via Google Vision labels) and in the text. Finally, named entities, locations, and dates were replaced with hypernyms (e.g., "Harrison Ford" replaced by "actor") to improve generalization. Human evaluation showed that over 90% of captions were judged as good.

The **Open Images V4** dataset [15] contains 9.2 million Creative Commons–licensed Flickr images with unified annotations for image classification, object detection, and visual relationship detection (see Figure 15). It is split into 9.0M training, 41K validation, and 125K test images. It provides 30.1 million image-level labels for 19.8k concepts, 15.4 million bounding boxes for 600 object classes, and 375k relationship annotations involving 57 classes. Each image contains on average eight annotated objects, reflecting the complexity of real-world scenes.

Compared to previous datasets such as MS-COCO or ImageNet, Open Images V4 offers more than 15 times as many bounding boxes and integrates multiple annotation types within the same images.

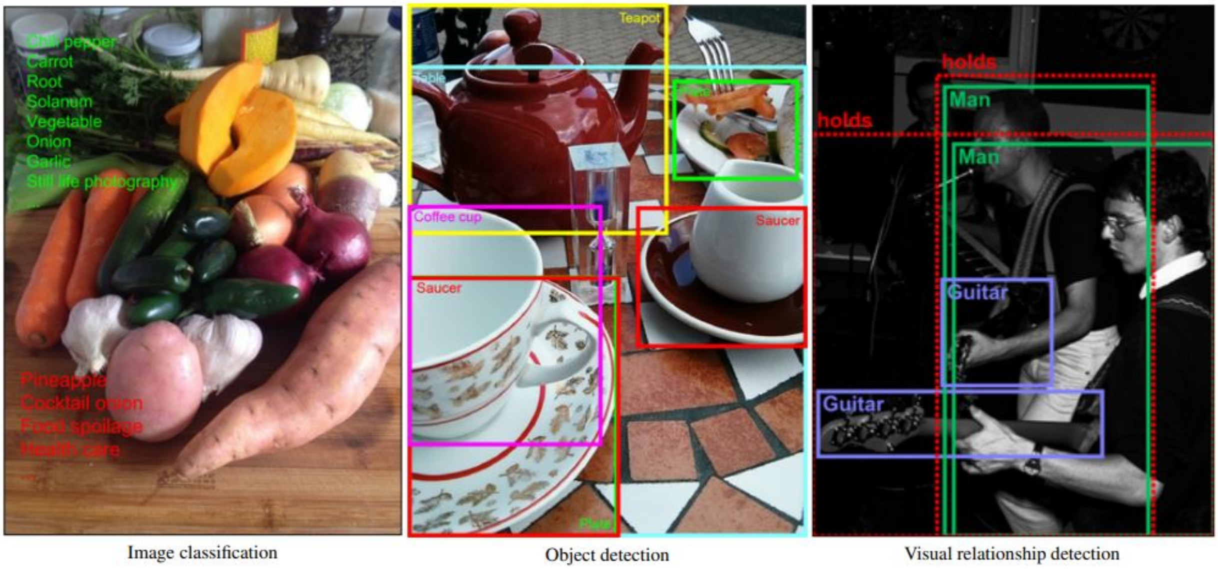

Figure 15: Example annotations from the *Open Images V4* dataset for image classification, object detection, and visual relationship detection. Green labels mark present classes, red labels absent ones. Dashed boxes group pairs of objects sharing a visual relationship.

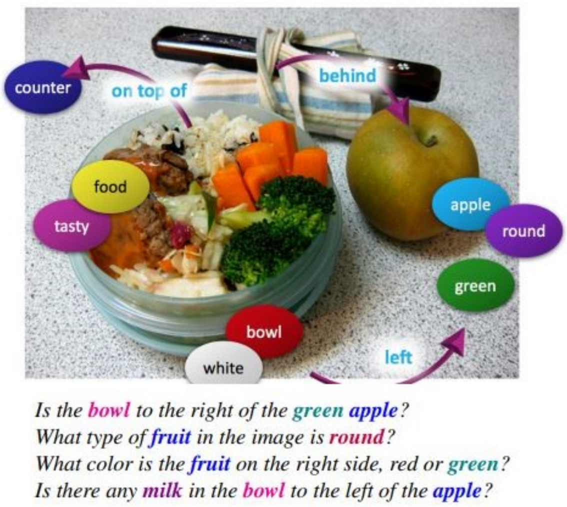

Figure 16: Example GQA images and queries illustrating spatial reasoning, attributes, and compositional question answering.

The **GQA** dataset [16] is built upon the **Visual Genome** scene graphs and contains 113k images with 22 million automatically generated yet semantically grounded questions, of which 1.7 million form a balanced training and evaluation split. Each question is associated with a *functional program* (a symbolic representation of its reasoning steps such as `select`→`filter`→`relate`→`query`) and with visual justifications linking words to specific image regions.

A distinctive feature of GQA is its emphasis on compositionality: a single question may reference nested visual relationships such as "the apple on the table to the left of the bowl." Human performance reaches 89.3%, while strong VQA systems achieve only around 54.1%, and a blind LSTM just 42.1%, illustrating the task's difficulty and the gap between recognition and reasoning. As the authors aptly remark, "It takes more than a smart guess to answer a good question," encapsulating the spirit of the GQA benchmark.

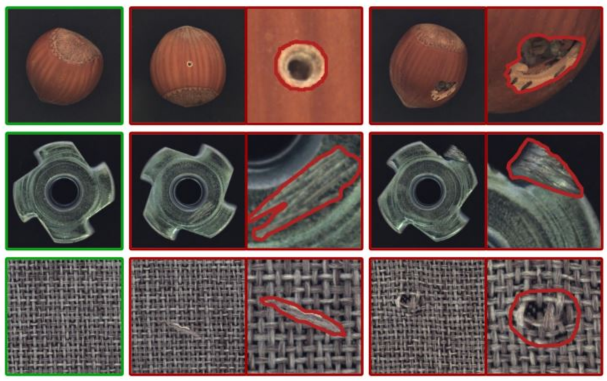

Figure 17: Examples from **MVTec AD**: each triplet shows a defect-free image (left), an anomalous image (middle), and the pixel-precise ground-truth mask overlaid on a zoomed region (right).

The **MVTec Anomaly Detection (MVTec AD)** dataset [17], created by the MVTec company, attempst to solve the issue that other datasets contain only normal (defect-free) imagery is available for training. It contains 5,354 high-resolution RGB/grayscale images spanning 15 categories (10 objects and 5 textures). Training splits comprise only normal images (3,629 total), while the test split (1,725 images) mixes normal and defective cases. Anomalies cover more than 73 defect types (e.g., scratches, dents, contaminations, missing parts, and structural deformations) and every defective image is annotated with pixel-accurate ground-truth regions (1,888 masks). Images were captured with industrial telecentric optics at $\approx 700 \times 700$–1024×1024 px (this means magnification of the image does not change with the object's distance from the lens), though some categories are grayscale only.

Figure 18: Example annotations from the **LVIS** dataset, illustrating fine-grained segmentation masks across diverse kitchen scenes.

The **LVIS** dataset [18] (*Large Vocabulary Instance Segmentation*) introduces over 1,000 entry-level object categories and aims to collect approximately two million high-quality segmentation masks over 164,000 images. Built with an "evaluation-first" philosophy, LVIS retains compatibility with the COCO average precision (AP) metric but dramatically increases category diversity while embracing the natural long-tailed distribution of real-world objects.

Unlike previous benchmarks, which exhaustively annotate a fixed set of categories, LVIS adopts a *federated* dataset design. Each category has its own positive and negative subsets of images that are exhaustively annotated for that specific class, avoiding the impractical burden of labeling every image with all categories. This federated approach ensures fairness during evaluation even when an object can belong to multiple overlapping or synonymous categories. As the authors explain, "a toy deer is both a toy and a deer," and the benchmark must fairly credit detectors that output either or both labels. Notably, LVIS segments were found to have higher boundary accuracy and annotator consistency than COCO and ADE20K masks, achieving over 99% acceptance after verification.

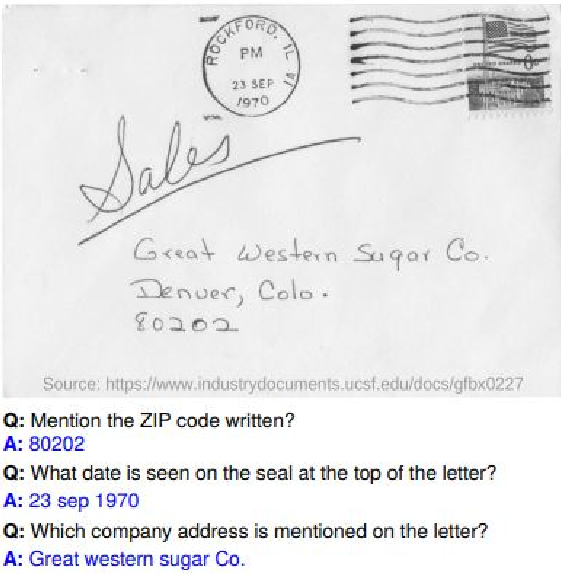

Figure 19: DocVQA example. Sample document image with Q&A pairs that require reading text and interpreting page layout/structure.

DocVQA [19] is a large-scale dataset for *visual question answering on document images.* It targets purpose-driven understanding of real documents (letters, forms, tables, lists, figures, and handwritten notes) to train systems that must both *read* text and *reason* over layout and structure. The dataset contains **12,767** pages drawn from **6,071** industry documents (spanning early 1900s–2018 and multiple domains) and **50,000** natural-language questions with one or more extractive answers.

Questions are answerable by copying spans *verbatim* from the page (an extractive QA setting). Success therefore couples robust OCR with structural understanding: models must leverage cues like headings, seals, key–value pairs, table cells, and relative position (e.g. "at the top left", "in the seal", "ZIP code"). Data are split **80/10/10** into train/val/test: 39,463/5,349/5,188 questions over 10,194/1,286/1,287 images. Human accuracy is reported at **94.36%**, leaving a substantial gap for current VQA/MRC baselines that struggle especially on structure-heavy queries.

The **DeepFake Detection Challenge (DFDC)** dataset [20] is a large, consented face-swap corpus designed for training and benchmarking Deepfake detectors. It comprises 100k+ 10-second clips sourced from 3,426 paid actors (average 14.4 videos/subject; mostly 1080p), totaling 38.4 days of footage and 25TB of raw data. Relative to previous datasets, DFDC is orders-of-magnitude larger, higher quality, and ethically collected with explicit consent.

Multiple techniques (Deepfake Autoencoder, morphable-mask, Neural Talking Heads, and so on) were used to make the fake videos so that detection models wouldn't just overfit to one kind of fake, but would learn to generalize to many.

The **TextCaps** dataset [21] introduces the task of image captioning with reading comprehension, where models must both recognize scene text and reason about it within the visual context. For example, recognizing a "red sign" is insufficient, the model must read "Mornington Crescent" to identify the location, or understand that "Rice is winning" implies interpreting a scoreboard rather than simply detecting numbers. The dataset contains 145,000 captions over 28,000 images belonging originally to OpenImages v3.

The **LAION-400M** dataset [22] contains an astounding 400 million image–text pairs, filtered using **CLIP** to ensure semantic alignment between images and captions. It was created as a community-driven effort to provide a large-scale, public alternative to proprietary datasets used to train models such as CLIP and DALL · E. Each entry includes an image URL, text caption, CLIP embeddings for both modalities, metadata (license type, image size, and NSFW tag), and precomputed kNN indices for efficient similarity search.

FairFace [23] introduces 108,501 in-the-wild face images with labels for race, gender, and age, constructed to be balanced over seven race groups: White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, and Latino, which were annotated manually.

The paper highlights that many widely used face datasets (e.g., celebrity or politician collections) are skewed toward lighter-skin/White faces, which leads to uneven accuracy across demographics. Models trained on FairFace yield higher and more consistent accuracy across race and gender on these novel sets, even when FairFace's training subset is down-sampled.

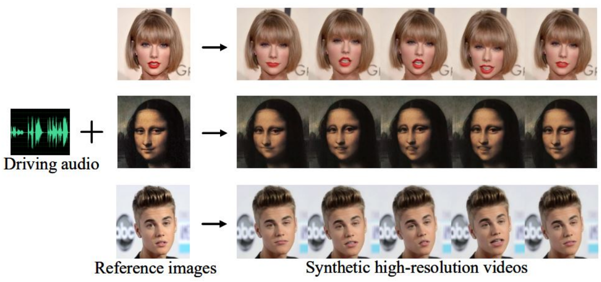

Figure 24: Overview of one-shot high-resolution talking face generation from the HDTF dataset. Given a reference image and driving audio, the model synthesizes high-resolution videos with synchronized facial animation.

HDTF [24] authors note that prior systems top out around 256×256, so that they introduce High-Definition Talking Face (HDTF), a high-definition, in-the-wild dataset (∼16 hours of 720p–1080p video, 300+ subjects, 10k+ sentences) curated from recent YouTube content so the crops remain sharp after 512×512 resizing. They synthesize high-definition, lip-synchronous videos (demonstrations include portraits and paintings) and report both objec-

Figure 20: DFDC face mosaic. Example frames from the DFDC dataset. All identities are consenting paid actors recorded specifically for creating and detecting face swaps.

Figure 21: Examples from TextCaps demonstrating how captions combine visual description with textual information in the image. Bold phrases indicate paraphrasing or reasoning; underlined ones are directly copied OCR text.

tive and user-study gains over landmark-based baselines, especially at 512×512.

The LAION-5B dataset [25] is the largest openly available image–text dataset ever released, designed to democratize research on large-scale multimodal models such as **CLIP**, **GLIDE**, and **Stable Diffusion**. It contains 5.85 billion CLIP-filtered image–text pairs collected from Common Crawl, including 2.32 billion English pairs, 2.26 billion multilingual pairs covering over a hundred languages, and 1.27 billion language-agnostic samples, which often describe products or places. Each entry is accompanied by metadata such as the image URL, caption text, image dimensions, cosine similarity between the image and text embeddings, and safety-related scores like watermark and NSFW probabilities.

The accompanying web interface allows interactive dataset exploration, nearest-neighbor search via CLIP embeddings, and flexible subset generation. The dataset has been validated through extensive experiments. Models trained on **LAION-400M** reproduce the zero-shot classification performance of OpenAI's CLIP on ImageNet and other benchmarks, confirming that the filtering and scale of LAION-5B provide data of comparable quality. Subsets such as LAION-Aesthetics and LAION-High-Resolution have also been derived to train models like Stable Diffusion, which achieve state-of-the-art results in text-to-image generation.

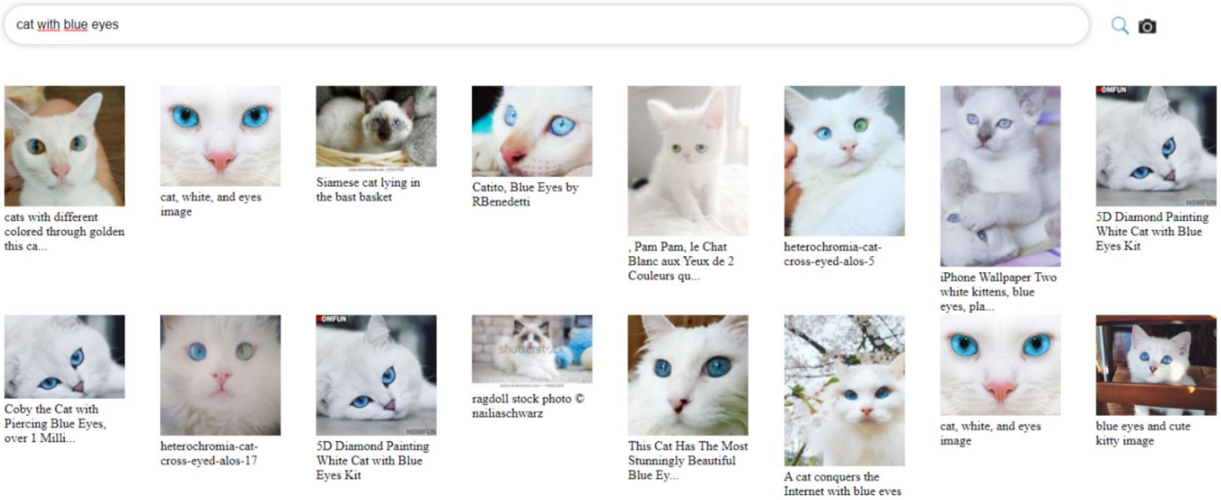

Figure 22: Example image retrieval results for the query "cat with blue eyes" using the LAION-400M web demo. The dataset supports large-scale multimodal similarity search using CLIP embeddings.

(a) FairFace (b) UTKFace (c) LFWA+ (d) CelebA

Figure 23: Example faces from (a) FairFace and (b–d) popular prior datasets. FairFace is designed to be racially balanced, unlike many existing sets that are dominated by White faces.

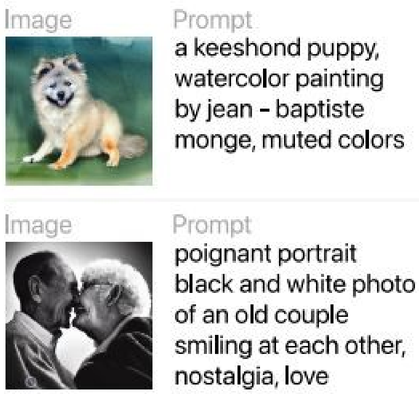

Figure 26: Examples from **DiffusionDB**, showing generated images with their corresponding prompts

DiffusionDB [26] is the first large-scale, community-generated prompt dataset for text-to-image diffusion models. It contains over 14 million images, 1.8 million unique text prompts, and detailed **Stable Diffusion hyperparameters** including seed, step count, CFG scale, sampler type, image size, and timestamp. The dataset was collected from the official Stable Diffusion public Discord server, where users shared their generation results. To ensure safe and ethical use, the authors applied state-of-the-art NSFW detection models on both images and prompts, producing continuous safety scores that allow flexible filtering. Prompts were also analyzed linguistically and semantically: most are short (6–12 tokens) and in English, but the dataset spans over 30 languages.

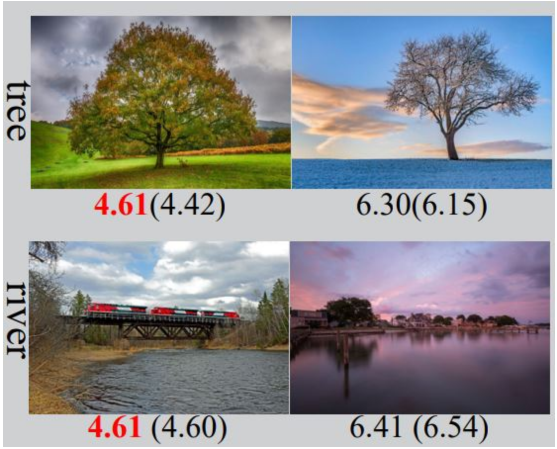

Figure 27: Examples from the **TAD66K** dataset [27]. Each pair shows images from the same theme with their average aesthetic scores (in red) and predicted scores in parentheses. Even when scores are similar, different themes correspond to distinct aesthetic criteria.

The **TAD66K** dataset [27] (Theme and Aesthetics Dataset) was created to address a key issue in image aesthetics assessment (IAA): different image themes follow distinct evaluation criteria, yet existing datasets and models generally ignore this fact. TAD66K contains approximately 66,327 carefully curated images covering 47 popular themes, organized into seven super-themes (plants, animals, artifacts, colors, humans, landscapes, and others). Each image was selected manually to ensure thematic clarity and diversity. Every image is annotated with an aesthetic score from 1 to 10, and unlike previous datasets that measure aesthetics, TAD66K provides theme-specific evaluation criteria. For instance, the aesthetic standards applied to images of trees differ from those for rivers or people.

MagicBrush [28] is a manually annotated dataset for instruction–guided real image editing. It contains 5,313 edit sessions and 10,388 turns (10K triplets: source image from MS COCO, instruction, target image). Images are modified with DALL · E 2 until an instruction-consistent, photorealistic target was obtained. As shown in Figure 28 some sessions are single-turn, others are multi-turn.

Figure 29: Pick-a-Pic web-app outputs. For each prompt the non-preferred image (left) is darkened and the user-preferred image (right) is highlighted.

Pick-a-Pic [29] is a dataset of human preferences for text-to-image generation, collected through a public web application in which users write prompts, generate images with modern diffusion models, and select which image they prefer (or declare a tie). Each datapoint contains a prompt, two model outputs, and a preference label. The initial release contains over $\sim$ 500K pairwise judgments drawn from $\sim$ 35K prompts (and continues to grow), with images produced by Stable Diffusion 2.1, Dreamlike Photoreal 2.0, and SDXL variants under multiple classifier-free guidance (CFG) scales.

InternVid [30] comprises ~7.1M public YouTube videos spanning ~760K hours and yields ~234M trimmed clips (median ~10 s) paired with ~4.1B caption words. Coverage is intentionally broad: 16 topical domains, ~6,100 action phrases used as queries, and geographically diverse sources; ~85% of videos are 720p (remainder 360–512p).

OpenVid-1M [31] is a large-scale open-scenario dataset containing over one million high-quality text–video pairs, each accompanied by very descriptive captions, as shown in Figure 31. Prior datasets such as WebVid-10M and Panda-70M prioritize size over fidelity and contain numerous low-resolution, watermarked, flickering, or blurry videos, such is the issue OpenVid-1M seeks to solve.

The dataset was built by processing large collections such as Panda-50M, ChronoMagic,

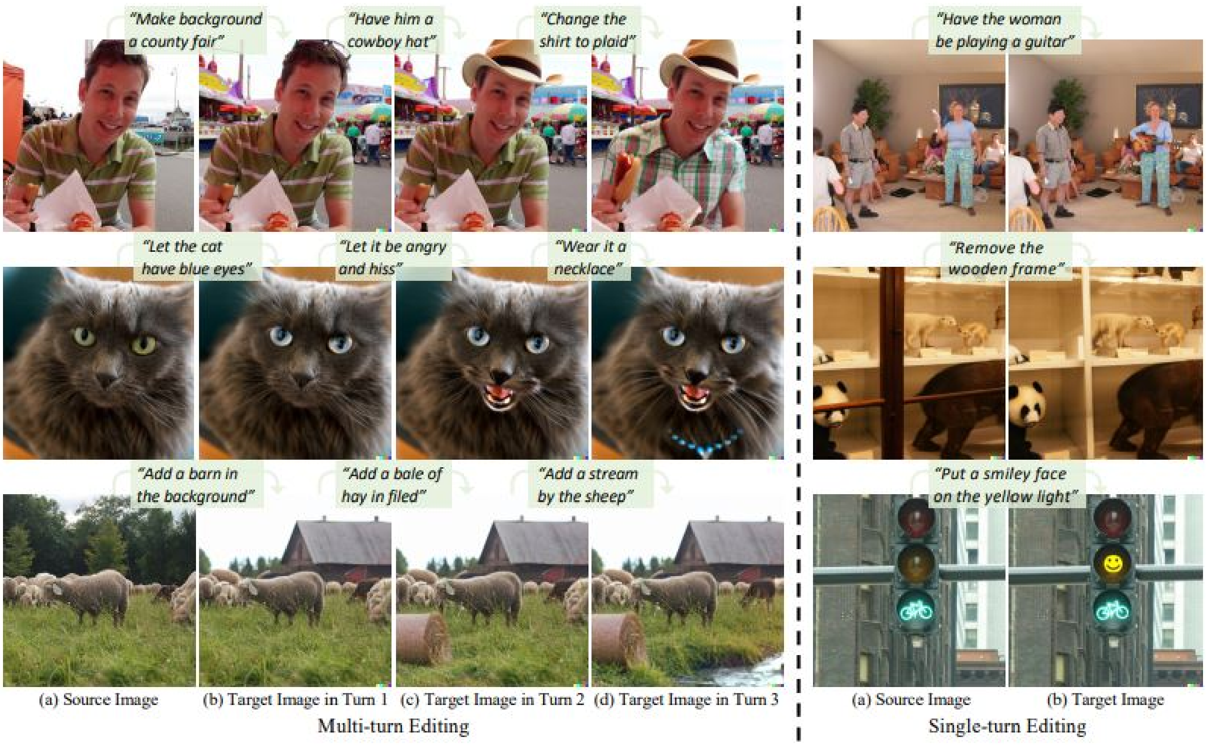

Figure 28: Illustration of MagicBrush: instruction–guided real image editing with both *multi–turn* (left) and *single–turn* (right) workflows. Each triplet consists of a source image, a natural–language instruction, and a target image.

CelebV-HQ, and Open-Sora-Plan. Videos were evaluated according to multiple criteria, such as using the LAION Aesthetics Predictor to retain only the top 20% of visually appealing clips. The rather descriptive captions were then generated or rewritten via the multimodal model LLaVA-v1.6-34B.

To further advance high-definition video generation, 433K of these clips at 1080p resolution were selected to create OpenVidHD-0.4M. Overall, OpenVid-1M comprises 1,019,957 clips averaging 7.2 seconds each (approximately 2,051 total hours of content), all at least 512 × 512 in resolution.

Experimental results confirm that OpenVid-1M and MVDiT significantly outperform prior approaches. Models trained on OpenVid-1M achieve higher aesthetic (VQAA) and technical (VQAT) scores, improved text–video alignment (SD_score and Blip_BLEU), and better temporal coherence (Clip_temp_score).

Visual CoT [32] contains 438K question–answer pairs, each annotated with a grounding bounding box that marks the critical image region required to answer the question; approximately 98K pairs additionally include step-by-step visual–textual reasoning. Data span five domains: text/doc understanding, fine-grained recognition, charts, general VQA, and relation reasoning, and are built from twelve source corpora (e.g., TextVQA, DocVQA, TextCaps, Flickr30k,GQA, Open Images).

HQ-Edit [33] comprises approximately 200,000 instruction-based image edits. Unlike earlier datasets that rely on people to give feedback, HQ-Edit collects data automatically using advanced models: GPT-4V and DALL-E 3. This method creates clearer, higher-quality images and more accurate text instructions, making the link between the text and image much stronger.

In addition to automatic evaluation, a human study on 1,651 image pairs generated by DALL-E 3 was conducted to validate the proposed metrics. Participants rated how well the output images reflected the corresponding edit instructions on a five-point scale, ranging from "totally unrelated" to "perfectly aligned." Pear-

Figure 30: Examples from **InternVid**. For each video clip (three frames shown), the dataset provides LLM-generated captions that emphasize nouns (blue) and verbs (green). Compared with raw ASR transcripts (grey), the captions are more visually grounded and descriptive.

son correlation analysis revealed that HQ-Edit's Alignment metric strongly correlates with human judgments and significantly outperforms the commonly used CLIP Directional Similarity score, which often fails to capture nuanced, instruction-level edits. Furthermore, fine-tuning InstructPix2Pix on HQ-Edit results in a substantial performance boost, achieving a 12.3-point increase in Alignment and a 5.64-point improvement in Coherence over the vanilla model.

BLIP3o-60k [34] is a 60,000-sample instruction-tuning dataset built by prompting GPT-4o with textual inputs covering various visual domains, including scenes, everyday objects, artistic styles, lighting conditions, gestures, and compositions. Each example in BLIP3o-60k consists of an instruction prompt, a corresponding image, and the image's CLIP feature embedding. The prompts are designed to mimic real-world creative instructions, such as "a person jumping in the rain wearing a yellow coat" or "an ancient city at sunset in watercolor style".

The **ImgEdit** dataset [35] contains 1.2M carefully curated edit pairs at high resolution (shorter side $\geq$ 1280), including 1.1M single–turn samples spanning 10 representative operations (local: add, remove, replace, alter, motion change, object extraction; global: background/style; visual edit via reference; hybrid edits that compose multiple local ops) and 110K multi–turn samples covering content memory, content understanding, and version backtracking.

The original images come from a high–aesthetics subset of LAION (aesthetic score $>$ 4.75 and resolution filtering). These go through multiple steps to generate the edits, including: the creation of concise captions and candidate edit targets with **GPT-4o**, validation of regions to be edited by CLIPScore and area thresholds (e.g., background edits must occupy $>$40% of the image), and prompt-guided post-filtering, returning quality scores.

ShareGPT-4o-Image [36] contains 91K synthetic samples comprising 45K text-to-image pairs and 46K instruction-guided text&image-to-image triplets, where every target image is produced by GPT-4o-Image. The two distinct sets go through specific pipelines to arrive at the result.

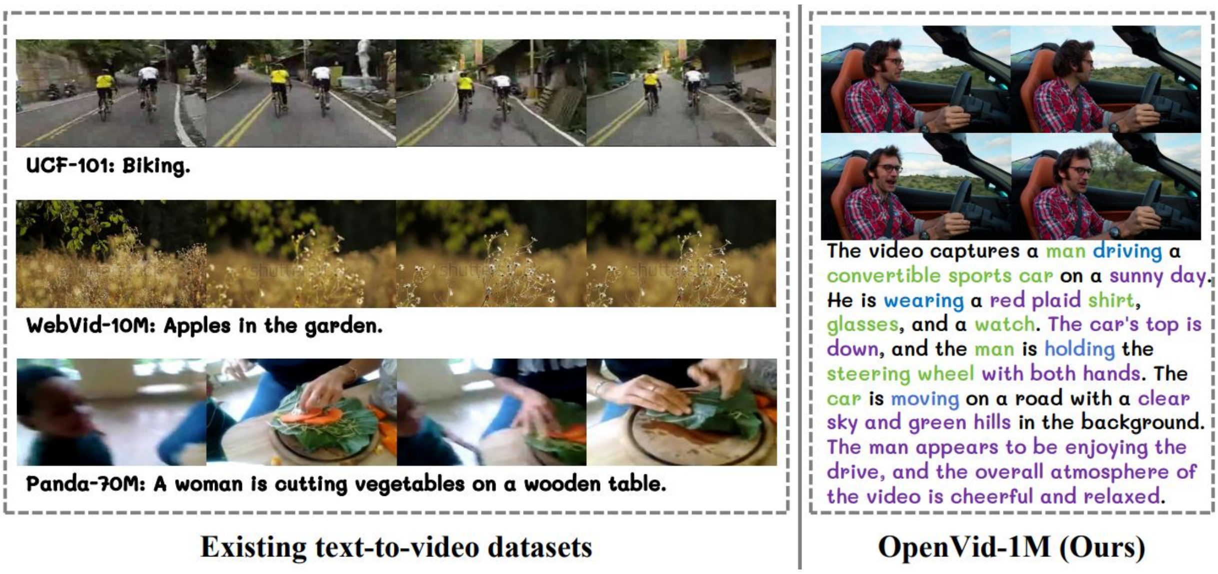

Figure 31: Comparison between existing text-to-video datasets and OpenVid-1M. Datasets such as UCF-101, WebVid-10M, and Panda-70M often contain low-resolution or low-quality videos with short captions, whereas OpenVid-1M features over one million high-quality clips with expressive, detailed textual descriptions highlighting both semantic and visual nuances.

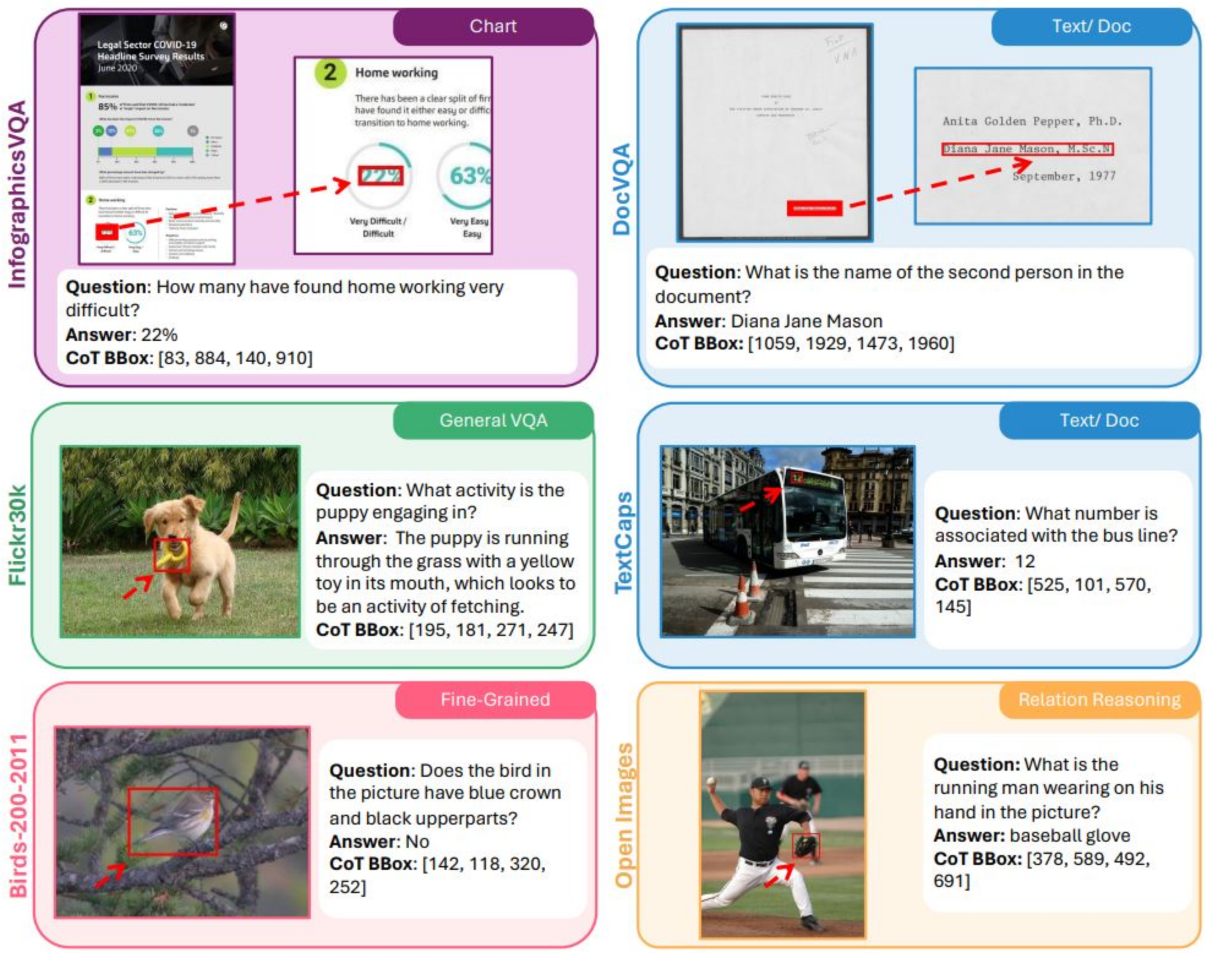

Figure 32: Visual CoT examples across five domains. Each question–answer pair is accompanied by an intermediate bounding box (red) that highlights the key region used in reasoning.

(a) Zoom in on the fox and add snowflakes falling around it.

(b) Alter her hair color to black.

(c) Replace the Amazon rainforest background with the underwater scenery of the Great Barrier Reef and adjust the parrot's position and wings to depict it flying.

(d) Replace the silver teapot with a ceramic blue and white patterned teapot with a similar wooden handle and a matching ceramic lid.

Figure 33: (a)–(d): example image edit pairs and instructions from HQ-Edit. (e): comparison of dataset quality between HQ-Edit and previous benchmarks. “Alignment” and “Coherence” are newly proposed metrics introduced in Section 3.4 to evaluate semantic and aesthetic quality.

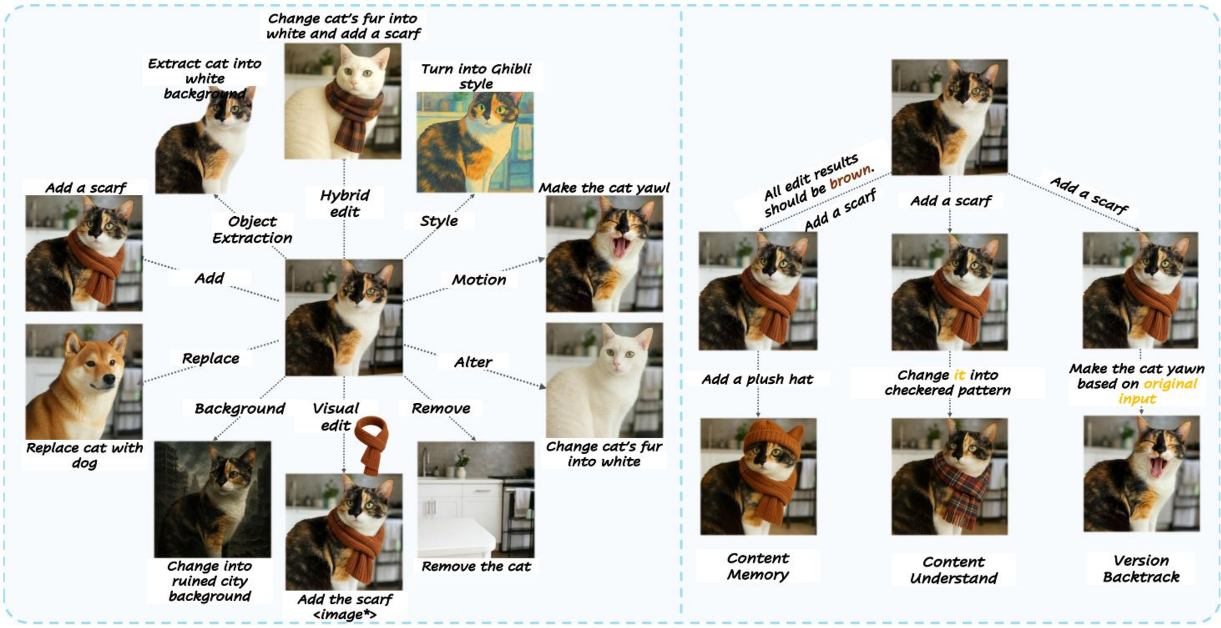

Figure 34: ImgEdit illustrates single–turn edits (add, remove, replace, alter, background/style change, motion change, object extraction, visual and hybrid edits) and multi–turn behaviors (content memory, content understanding, version backtracking).

Figure 35: ShareGPT-4o-Image. Examples from the two splits: *45K* text-to-image (left) and *46K* text&image-to-image (right) synthesized by GPT-4o-Image.